Chanyoung Moon

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RESEARCH INTERESTS

Computer Architecture, Hardware–Software Co-design, and Domain-Specific Accelerators

EDUCATION

- **University of Wisconsin–Madison** *Sep. 2025 – May 2026*
  Visiting Exchange Student, Electrical and Computer Engineering Madison, WI, USA
  - GPA: 3.94/4.0
- **Korea University** *Mar. 2021 – Aug. 2027 (Expected)*
  B.E. in Electrical Engineering Seoul, South Korea
  - Overall GPA: 4.44/4.5 (Upper-Division: 4.50/4.5), Major GPA: 4.46/4.5
  - Leave of absence for mandatory military service: Apr. 2023 – Oct. 2024

PUBLICATIONS

[C.1] Chanyoung Moon, Travis Pence, Daisuke Yamada, Jiaqi Mo, Yanxin Zhang, Vikas Singh, Karthikeyan Sankaralingam. **FORGE: Procedural Weight Generation for Sparse MoE Inference** [paper]. ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2027).

[C.2] Travis Pence, Daisuke Yamada, Jiaqi Mo, Chanyoung Moon, Karthikeyan Sankaralingam, Vikas Singh. **Fast Sandwich Products in Clifford Algebra**. Advances in Neural Information Processing Systems (NeurIPS 2026).

[C.3] Chanyoung Moon, Hyung Jin Cho, Junil Kim, Seon Wook Kim. **Study for Behavior Verification and Resource Usage Analysis of RISC-V R-Type ISAs on FPGA**. IEIE Fall Conference 2022 (Korea).

RESEARCH EXPERIENCE

- **Vertical Research Group, University of Wisconsin–Madison** *Oct. 2025 – Present*
  Undergraduate Researcher (Advisor: Prof. Karthikeyan Sankaralingam, also at NVIDIA) Madison, WI, USA
  - **Closed-Loop Neural–Machine Interface System Simulation**
    Leading simulator development for a simulation-first BCI/exoskeleton co-design project; modeled the sensorimotor loop as a multi-rate dependency graph under a sub-100 ms budget to identify architectural bottlenecks.
  - **3D Gaussian Splatting Rendering Acceleration**
    Reproduced GEMM-GS (DAC 2026) as a baseline, profiled it on H100, and identified the sequential alpha-blending stage as the dominant remaining bottleneck; prototyping parallelization of its dependency chain on tensor cores.
  - **Fast Sandwich Products in Clifford Algebra**
    Contributed to the GPU kernel for the rotor sandwich product with bank-conflict-free shared-memory access, yielding up to 64× wall-clock and 35× memory improvements over existing implementations ([**C.2**]).
  - **FORGE: Procedural Weight Generation for Sparse MoE Inference**
    - Led the project and wrote the first-author paper ([**C.1**]).
    - Designed the Rotor Generation Unit (RGU), an on-SM accelerator that regenerates MoE expert weights on the fly, trading HBM weight traffic for on-chip computation; exposed three axes of data parallelism in the generation algorithm and mapped them onto the datapath.
    - Implemented and verified the RTL in SystemVerilog with cocotb and Verilator, synthesized for area and timing, and built a system-level simulator to project end-to-end speedup.
    - Integrated RGU into Accel-Sim for cycle-level GPU simulation after submission.
    - Cut parameter memory by 28–29% and raised system throughput 1.3–2.8× at ∼0.9% GPU SM area (7 nm).

- **Computer Systems Laboratory (CSL), Korea University** *Sep. 2026 – Dec. 2026 (Expected)*
  Undergraduate Thesis (Advisor: Prof. Jeongseob Ahn) Seoul, South Korea
  - Investigating harness-aware KV cache management for agentic LLM serving: using agent-harness signals (e.g., subagent return, context compaction) to evict dead KV blocks early in vLLM with CPU-tier offloading.

- **Compiler & Microarchitecture Lab (COMMIT), Korea University** *Jul. 2022 – Sep. 2022*
  Undergraduate Research Intern (Advisor: Prof. Seon Wook Kim) Seoul, South Korea
  - Implemented and verified RISC-V R-Type instruction modules in Verilog HDL on FPGA; first-authored **[C.3]**.

HONORS AND AWARDS

- **Minister of Trade, Industry and Resources Award** *Sep. 2026*
  1st place among 213 scholars; Korean government award for outstanding academic and research achievement through the Korea–U.S. Advanced Fields Youth Exchange Program
- **NRF Undergraduate Research Fellowship (∼$4,500)** *Sep. 2026 – Feb. 2027*
  National Research Foundation of Korea (for outstanding STEM undergraduates pursuing research careers)
- **3rd Korea–U.S. Advanced Fields Youth Exchange Scholarship (∼$18,000)** *2025 – 2026*
  Ministry of Trade, Industry and Resources, Republic of Korea
- **Endowment Scholarship (∼$3,700, Awarded to Top Student)** *May 2025*
  School of Electrical Engineering, Korea University
- **Academic Excellence Scholarship (∼$11,200 total)** *Spring 2022, Spring 2025, Fall 2025*
  Korea University (awarded for 4.5/4.5 GPA)
- **Guwon Scholarship Foundation Award (∼$1,800)** *2022*
  For academic excellence
- **Dean's Award** *Fall 2021, Fall 2022, Spring 2025*
  Korea University (awarded for 4.5/4.5 semester GPA)
- **Semester Highest Honors** *5 semesters, 2021 – 2025*
  Korea University

OTHER EXPERIENCE AND ACTIVITIES

- **Architecture & System for AI Seminar, HandS Academic Society, Korea University** *Fall 2026*
  - Co-hosted the seminar; gave five lectures to 21 students on GPU architecture, HBM, and interconnects.
- **Korean Language Buddy, Korean Language Center, Korea University** *Sep. 2026 – Nov. 2026*
  - Tutored two international students (English- and Japanese-speaking) in Korean for 30+ hours.
- **Defense Communication Command, Republic of Korea Army** *Apr. 2023 – Oct. 2024*
  Microwave Communications Operator (Squad Leader) Daejeon, South Korea
  - Served as Squad Leader; participated in two ROK–US combined forces exercises (UFS 2023, 2024).
- **Young Engineers Honor Society (YEHS)**, Seoul, South Korea *May 2025 – Present*
  - Dean-recommended honor society of the **National Academy of Engineering of Korea (NAEK)**.
- **Woonhwa Club, Korea University** *Apr. 2021 – Aug. 2022*
  Vice President, 53rd Cohort Seoul, South Korea
  - Taught weekly science classes to middle-schoolers from underserved households and coordinated the tutoring program.

SKILLS

- **Programming & Hardware Description:** C, C++, CUDA, Python, Verilog, SystemVerilog
- **EDA & Verification:** Synopsys DC, Yosys, Vivado, Quartus, ModelSim, cocotb, Verilator, Icarus Verilog
- **Performance Analysis & Simulation:** Accel-Sim, CACTI, NVIDIA Nsight Compute (NCU)